

WHAT MAKES A POST GO VIRAL?



**A CROSS-
PLATFORM
DATA
ANALYSIS**



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PROBLEM STATEMENT

WHAT POST CHARACTERISTICS ARE ASSOCIATED WITH VIRAL CONTENT ON SOCIAL MEDIA PLATFORMS?



Understanding virality helps explain engagement behaviour and supports better content strategies. By analysing post metadata and engagement metrics across Facebook, Instagram, and TikTok, we aim to identify common patterns linked to viral posts.

DATA SOURCES

OVERVIEW

To study post characteristics and virality, we use three complementary social media datasets, each representing a different platform and content format.

These datasets provide post-level metadata and engagement metrics, allowing both platform-specific and cross-platform analysis.



FACEBOOK POSTS DATASET



- **TEXT-BASED POSTS**
- **INCLUDES POST METADATA AND ENGAGEMENT INDICATORS**
- **USEFUL FOR ANALYSING VIRALITY IN TRADITIONAL SOCIAL MEDIA FORMATS**

INSTAGRAM POSTS DATASET

- IMAGE-BASED CONTENT
- CONTAINS CAPTION, HASHTAG USAGE, AND ENGAGEMENT METRICS
- CAPTURES VISUALLY-DRIVEN ENGAGEMENT BEHAVIOUR



TIKTOK POSTS DATASET



TikTok

- **SHORT-FORM VIDEO CONTENT**
- **INCLUDES VIDEO DURATION, POSTING TIME, HASHTAGS, AND ENGAGEMENT METRICS**
- **REPRESENTS FAST-PACED, ALGORITHM-DRIVEN VIRALITY**



DATA CHARACTERISTICS & INITIAL EXPLORATION

DATA CHARACTERISTICS

Across all three platforms, the datasets contain two key components:

POST METADATA

- Posting time (date, hour, day of week)
- Content descriptors (text captions, hashtags)
- Media attributes (e.g. video duration for TikTok)

ENGAGEMENT METRICS

- Views / impressions
- Likes
- Shares
- Comments

PROBLEM FORMULATION & VIRAL DEFINITION

PROBLEM FORMULATION

How do post characteristics influence the likelihood of a post becoming viral across social media platforms?

VIRAL DEFINITION

Virality is approximated using engagement-based thresholds:

- Facebook & Instagram: high engagement (likes, comments, shares)
- TikTok: views $\geq 1,000,000$

This binary definition enables consistent comparison across platforms and handles highly skewed engagement data.



DATA PREPARATION & FEATURE SELECTION

- Selected relevant post metadata and engagement metrics
- Merged datasets at the post level
- Checked for missing and inconsistent values

KEY FEATURES

- Posting time
- Content attributes (duration, hashtags)
- Engagement metrics (views, likes, comments, shares)

DATA QUALITY & LIMITATIONS

DATA QUALITY CONSIDERATIONS

- Engagement metrics are clearly defined and consistently available
- Post-level metadata is sufficient for exploratory analysis

LIMITATIONS

- Engagement values are snapshot-based and do not reflect the speed of virality
- Available features vary across platforms

SUMMARY & NEXT STEPS

SUMMARY:

- Defined a clear data science problem focused on post virality
- Identified relevant and complementary datasets across platforms
- Conducted initial data profiling and feature selection
- Confirmed dataset suitability for further analysis

NEXT STEPS:

- Perform detailed data cleaning and feature engineering
- Conduct exploratory data analysis (EDA)
- Build and evaluate predictive models for virality

GITHUB & MEMBERS

GITHUB LINK

<https://github.com/SlowToes/SC3021-SDAD-DS.git>

MEMBERS

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THANK YOU!